Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/27/2006
PI Name:	Macknight, Allen Ph.D.		
Project Title:	Wastewater Processing Cascade Distillation Subsystem		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	al.macknight@honeywell.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	310-512-3307
Organization Name:	Honeywell		
PI Address 1:	Honeywell 2525 W. 190th Street		
PI Address 2:	Mail Stop 36-1-93120		
PI Web Page:			
City:	Torrance	**State:**	CA
Zip Code:	90504-6099	**Congressional District:**	36
Comments:	alex.lubman@honeywell.com		
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/01/2004	**End Date:**	12/31/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HE01A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project will develop and demonstrate an advanced wastewater processing subsystem that is based on centrifugal vacuum distillation, and it will be conducted by a Rapid Technology Demonstration Team comprised of commercial, government, and academic partners. The Wastewater Processing Cascade Distillation Subsystem, hereinafter referred to as Cascade Distillation Subsystem (CDS), utilizes an innovative and proven multi-stage thermodynamic process to efficiently produce purified water, and its rotary centrifugal design provides gas/liquid phase separation and liquid transport (pumping) under microgravity conditions. Other key design features, such as the use of mist separators, hydraulic seals, and pitot pumps result in a spacecraft wastewater processor design that is superior to previously developed distillation processes. The objective of the proposed program is to advance the Technology Readiness Level of the Cascaded Distiller Subsystem from its current status of TRL 4 to a TRL of 6 by testing the subsystem in a relevant environment. This will be accomplished by testing the subsystem with real human-generated wastewater		

	sources in the NASA JSC Advanced Water Recovery System Development Facility, and verifying the microgravity compatibility of the subsystem through analysis Previous work has shown that the core components of the CDS can process a wide spectrum of wastewater sources in a relevant environment (Technology Readiness Level 4). Depending on the mission scenario, it is expected that the CDS can be optimized to function as a primary, secondary, or brine processor, or as a dedicated urine processor. . Preliminary equivalent mass analysis indicates that the CDS has the lowest equivalent system mass of known water processors by nearly a factor of 2. Successful completion of this project will advance the technology readiness level of this subsystem and provide NASA with sufficient information to evaluate the continued development of the CDS into a mature flight test project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated:)